

Variance Components from Nuclear Family Trios: A Linear Mixed Model Implementation in R

1 Introduction

Quantitative genetic variance can be decomposed into additive genetic (A), dominance genetic (D), shared environmental (C), and unique environmental (E) components.

Twin studies traditionally estimate ACE or ADE models, but many modern cohorts consist of nuclear family trios (mother–father–child). This paper presents a linear mixed model framework for estimating AE, ACE, and ADE models from trio data using nlme in R.

2 Variance Decomposition

For an individual phenotype y ,

$$y = \mu + a + d + c + e$$

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_C^2 + \sigma_E^2$$

3 General Pedigree Covariance

The covariance between relatives i and k under the ACDE model is

$$\text{Cov}(y_i, y_k) = 2\phi_{ik}\sigma_A^2 + \Delta_{ik}\sigma_D^2 + \gamma_{ik}\sigma_C^2$$

- ϕ_{ik} = kinship coefficient (additive sharing)
- Δ_{ik} = probability of sharing two alleles IBD (dominance)
- γ_{ik} = indicator of shared family environment

4 Coefficients for Nuclear Family Trios

Pair	2ϕ	Δ	γ
Mother–Father	0	0	1
Mother–Child	$\frac{1}{2}$	0	1
Father–Child	$\frac{1}{2}$	0	1

4.1 Key Consequence

Dominance deviations are not transmitted from parents to offspring:

$$\Delta_{ik} = 0 \quad \text{for all trio pairs}$$

Therefore dominance contributes only to individual variance:

$$\text{Var}(y) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2$$

and produces no observable covariance between relatives.

5 Identifiable Family Resemblance

Trio data contain only one source of familial covariance:

$$\text{Cov}_{family} = \frac{1}{2}\sigma_A^2 + \sigma_F^2$$

where σ_F^2 is an unidentified family resemblance component.

This component can be interpreted as:

$$\sigma_F^2 = \begin{cases} \sigma_C^2 & \text{ACE interpretation} \\ \text{residual familial resemblance} & \text{ADE interpretation} \end{cases}$$

Hence ACE and ADE models are mutually exclusive in trio data.

6 Mixed Model Representation

We model stacked trio observations:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}_A\mathbf{a} + \mathbf{Z}_F\mathbf{f} + \boldsymbol{\varepsilon}$$

$$\mathbf{a} \sim N(0, \sigma_A^2 \mathbf{A}) \tag{1}$$

$$\mathbf{f} \sim N(0, \sigma_F^2 \mathbf{I}) \tag{2}$$

$$\boldsymbol{\varepsilon} \sim N(0, \sigma_E^2 \mathbf{I}) \tag{3}$$

7 Data Restructuring in R

```
library(nlme)
```

```

longdata <- reshape(
  widelong,
  varying = c("bmi1","bmi2","bmi3"),
  v.names = "bmi",
  timevar = "member",
  times = c("mother","father","child"),
  idvar = "famid",
  direction = "long"
)

longdata$Acoef <- ifelse(longdata$member=="child",1,0.5)
longdata$Dcoef <- 1

```

8 AE Model

$$Var(y) = \sigma_A^2 + \sigma_E^2$$

```

model_AE <- lme(
  bmi ~ 1,
  random = list(
    famid = pdDiag(~ Acoef - 1),
    indid = pdDiag(~ Dcoef - 1)
  ),
  data = longdata,
  method = "REML"
)

```

9 ACE Model

$$Var(y) = \sigma_A^2 + \sigma_C^2 + \sigma_E^2$$

```
model_ACE <- lme(
  bmi ~ 1,
  random = list(
    famid = pdBlocked(list(
      pdDiag(~ Acoef - 1),
      pdDiag(~ Dcoef - 1)
    )),
    indid = pdDiag(~ Dcoef - 1)
  ),
  data = longdata,
  method = "REML"
)
```

10 ADE Model

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2$$

In trio data the ADE model estimates additive genetic variance and a combined non-additive/unique environmental component.

```
model_ADE <- lme(
  bmi ~ 1,
  random = list(
    famid = pdDiag(~ Acoef - 1),
    indid = pdDiag(~ Dcoef - 1)
  ),
  data = longdata,
  method = "REML"
)
```

```
data = longdata,  
method = "REML"  
)
```

11 Extracting Variance Components

```
VarCorr(model_ACE)
```

```
VarCorr(model_AE)
```

```
VarCorr(model_ADE)
```

12 Model Comparison

```
anova(model_AE, model_ACE)
```

```
anova(model_AE, model_ADE)
```

13 Interpretation

- AE vs ACE tests shared environmental effects.
- AE vs ADE tests non-additive genetic effects.
- ACE vs ADE cannot be fit simultaneously.

14 Conclusion

Nuclear family trios allow estimation of additive genetic variance and a single familial resemblance component. This component may be interpreted as shared environment (ACE) or as residual familial resemblance (ADE). Model comparison provides evidence for the most plausible interpretation.